PATIENT CARE THEORY 2

UNIT 7, PART 2: Paediatric Breathing

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Almost all pediatric “codes” are of respiratory origin

Sick pediatric patients are a low-volume, high-acuity call

How Does An Immature Nervous System Predispose To Hypoxia?

Parasympathetic tone

- hypoxia causes bradycardia
- bradycardia causes hypotension
- bradycardia worsens hypoxia

Scared children cry

- ↑ secretions
- ↑ airway irritability
- ↑ risk laryngospasm
- ↑ risk wheezing
- ↑ airway edema

Children easily sedated

- immature neurons
- less myelinization
- weak blood–brain barrier
- little prior exposure to drugs
- ↑ risk respiratory depression

Children uncooperative

- can't follow instructions
- struggle and fight
- stress ↑ metabolic rate

Apnea of prematurity

- the more premature, the greater the risk
- altered ventilatory responses to:
  - hypoxia
  - hypercarbia
  - sleep
- can be made worse if the infant is stressed, cold or ill
- apnea → hypoxia → bradycardia → worse hypoxia

Breathing

- Small airway passages obstruct easily
- Horizontal ribs, weak accessory muscles = Poor respiratory reserve – poor chest compliance (belly breathing)
- Swallowed air may limit ventilations
- **Anticipate need to assist ventilation**

Breathing

- Fast breathing may be normal
- Breathing at normal adult rates (10-20/min) may indicate respiratory failure in an infant/child
- Normal rate?
- Auscultation of chest may be misleading (transmitted breath sounds)
- Auscultation can be effective even when the child is crying

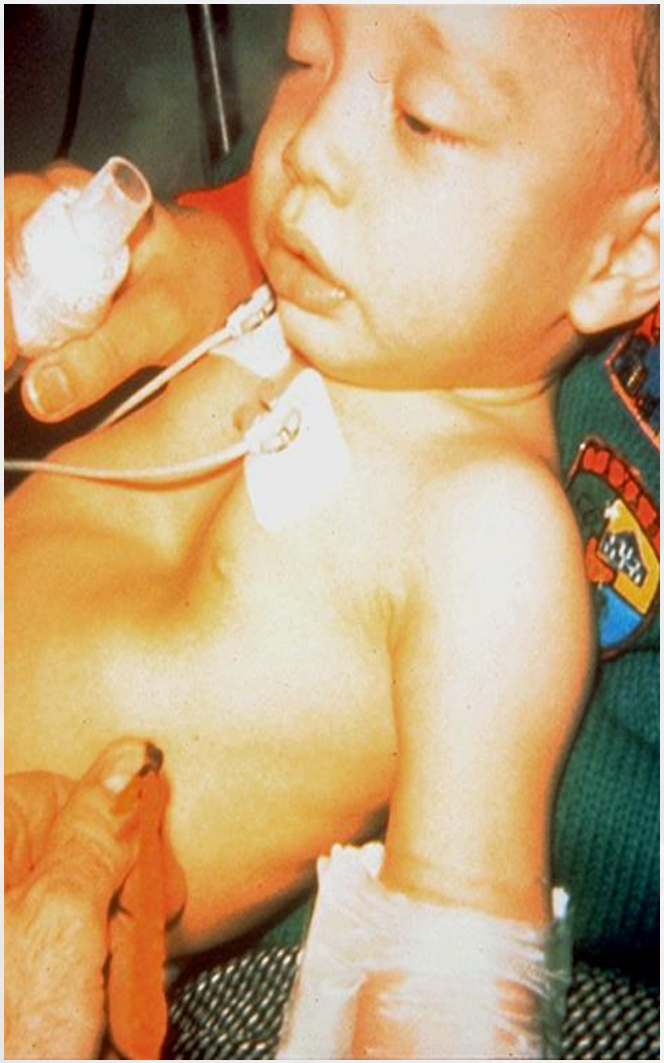
Breathing

- Higher metabolic demand = higher demand for O_2
- High metabolic rates + Low reserve capacity = High sensitivity to airway, breathing problems
- **Oxygenate, ventilate aggressively (but do not overventilate!)**

Too much ventilation can pressure vessels in chest, reducing venous return and cardiac output

Assessment Clues

- Fatigue = impending or actual respiratory failure
- Cyanosis
  - O2 sats; limitations?
  - Peripheral vs central



Aspiration

Etiology

- inhalation of either oropharyngeal or gastric contents into the apices
- Altered mental status
  - e.g. Stroke, head injury,
- Inability to protect own airway – e.g. loss of gag reflex

Sequelae

- Pneumonitis, pneumonia
- Aspiration of small amounts of oropharyngeal contents over a prolonged period of time leads to an infectious process

Aspiration

Presentation

- Onset of distress can be ≤ 1 hour
- Dyspnea
- Tachypnea, tachycardia
- +/- Fever (depending on how recent the event)
- Cyanosis
- Crackles: short popping sounds heard during inspiration in the early stages (often apical rather than basilar)
- ↓ breath sounds
- Pleural friction rub

Asthma

Etiology

- ***Chronic airway inflammation***
- ***Bronchial hyperreactivity***
- ***Bronchospasm*** (“reversable airway obstruction”)
- Mucous production
- Exacerbations: exercise, infection, cold/heat, smoke, pollen, atmospheric pollutants, medication non-compliance (extrinsic asthma)
- Non-immunologic (intrinsic asthma): reactions occur without the above precipitants

Asthma

Presentation

- ***Dyspnea***
- ***Cough***
- ***Wheezing***

Classic triad

- Tachypnea, tachycardia
- Chest tightness
- Accessory muscle use, nasal flaring
- Air trapping (↓↓↓ A/E severe cases), chest hyperinflation
- Pulsus paradoxus

Status Asthmaticus

Presentation

- **Altered LOA**
- **≤ 3 word dyspnea**

Status Asthmaticus

Presentation

- acute exacerbation unresponsive to initial treatment with bronchodilators
- carbon dioxide retention, hypoxemia, and respiratory failure
- persistent wheezing with retractions
- may present with cough, dyspnea, or emesis
- Silent chest**
- Lethargy (↓ CNS signifies exhaustion and respiratory failure)
- *status asthmaticus can lead to pneumothorax

Cystic Fibrosis

Etiology

- Genetic disorder
  - Abnormal chloride ion transport on surface of epithelial cells in exocrine glands leading to viscid (sticky) secretions
  - Defective: Cystic fibrosis transmembrane conductance regulator (CFTC) is channel/enzyme which mediates passive diffusion of chloride and bicarbonate through epithelial cell membranes

Cystic Fibrosis

Etiology

- Abnormally thick & tenacious secretions
- mucous
- Obstructs small bronchi and bronchioles – air trapping
- Stagnant mucous leads to recurring infections
  - Respiratory infections are the most prominent manifestation of CF (before 1 yr of age)
- Other organs also affected: e.g. pancreas, intestine
  - Thick secretions block the pancreatic ducts; digestive problems, IDDM, liver disease
- 1: 3,500 birth

Etiology

- Excess salt (Cl) excretions can led to electrolyte imbalances
  - Dysrhythmias
  - Heat exhaustion (environment may not seem “hot”; worse in younger children)
  - Increased risk of shock
- Trauma – malabsorption of Vit K
  - Lead to coagulation issues
  - Osteoporosis (??)

Cystic Fibrosis

Presentation

- Mom may notice salty skin on infant at birth
- Dyspnea
- Chronic cough + frequent lung infection
- Thick secretions / sputum
- Digestive issues
- Fractures

FEBRILE RESPIRATORY ILLNESS

Bronchiolitis

Pneumonia

Bronchiolitis

Etiology

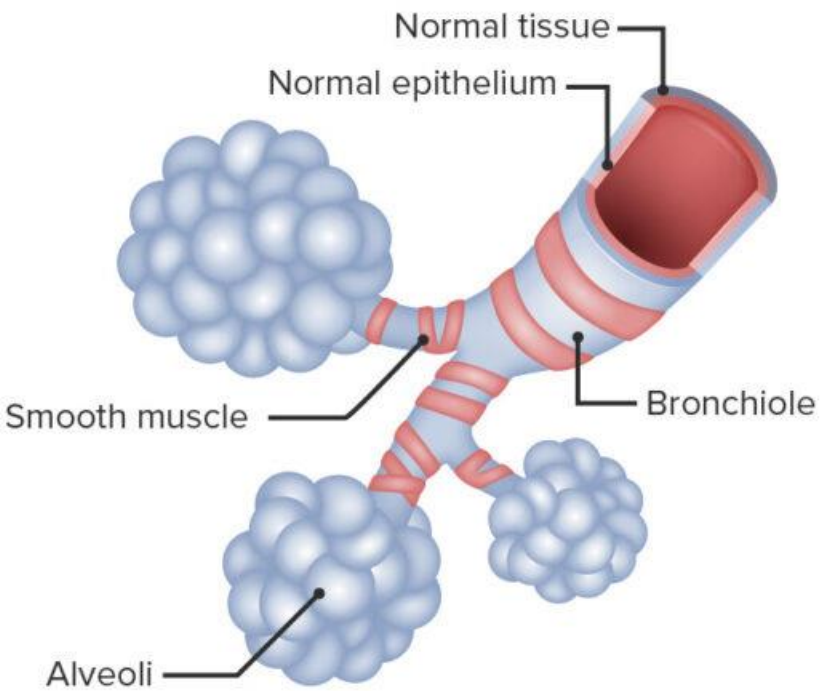
- Inflammation of small airways in the lower respiratory tract (bronchi and bronchioles) – may lead to necrosis
- Respiratory syncytial virus (RSV)
- Transmitted by oral droplet – often in late fall and winter months
- Common in children 2-12 months (<2 yrs)

Bronchiolitis

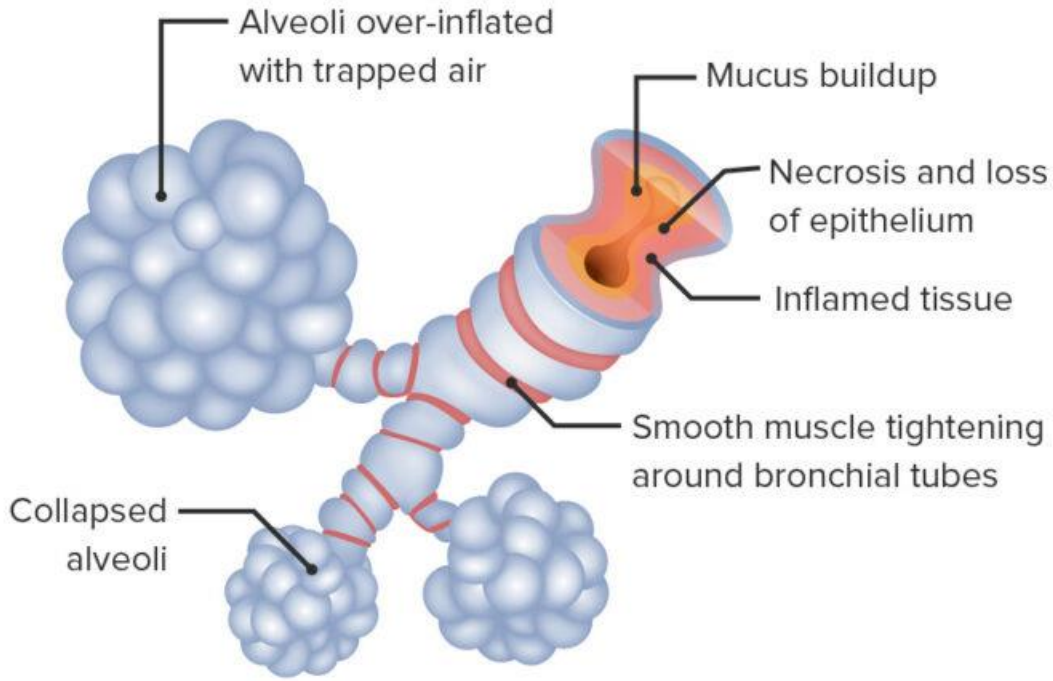
Presentation

- ***Recent fever, cold***
- ***Dyspnea***
- ***Wheezing***
- Crackles
- Chest retraction

Bronchiolitis pathophysiology



Normal bronchial tubes



Bronchial tube during bronchiolitis

Pneumonia

Etiology

- Virus, ***bacterium*** or fungus
- Common: Streptococcus pneumoniae, Haemophilus influenza, Mycoplasma pneumoniae, aspiration, etc
- Infection of lung parenchyma
  - Inflammation
  - Alveolar exudate

Pneumonia

Presentation

- ***Fever***
- ***Dyspnea***
- Tachypnea
- ***Cough (with or without sputum)***
- Tachycardia
- ***Pleuritic chest pain***
- Diaphoresis, fatigue, malaise
- Auscultation: bronchovescicular sounds over consolidation, ↓ A/E, +/- unilateral crackles
- Pleural friction rub

MANAGEMENT

Dyspnea or SOB

- Position for comfort – minimize activity
- Oxygen therapy
- PPV prn
  - If signs of respiratory failure – e.g. ≤ 3 word dyspnea, ↓ LOA,

Wheezing

- Salbutamol (Ventolin)
- Sympathomimetic, bronchodilator
- Predominantly a beta$_2$ agonist
  - bronchodilation
  - Also a beta$_1$ agonist which increases heart rate
  - Repeated doses = increased beta$_1$ effects
- Nebulizer or MDI and aerochamber

Bronchoconstriction – ALS PCS 5.3

- Indications:
  - Respiratory distress AND
  - Suspected bronchoconstriction
- Conditions:

can't ventilate until they're openeed up, cause it can trap air and cause pneumothorax

Salbutamol	
Age	N/A
LOA	N/A
HR	N/A
RR	N/A
SBP	N/A
Other	N/A

EPINEPHrine	
Age	N/A
Weight	N/A
LOA	N/A
HR	N/A
RR	BVM ventilation required
SBP	N/A
Other	Hx of asthma

Dexamethasone	
Age	N/A
LOA	N/A
HR	N/A
RR	N/A
SBP	N/A
Other	Hx of asthma OR COPD OR 20 pack-year history of smoking

Bronchoconstriction– ALS PCS 5.3

- Contraindications

Salbutamol
Allergy or sensitivity to salbutamol

EPINEPHrine
Allergy or sensitivity to EPINEPHrine

Dexamethasone
Allergy or sensitivity to steroids
Currently on PO or parenteral steroids

**PO and parenteral steroids refer to those that are systemic*

Bronchoconstriction - Treatment– ALS PCS 5.3

Consider salbutamol				
	Weight <25 kg		**Weight** ≥25 kg	
	Route MDI*	**Route** NEB	**Route** MDI*	**Route** NEB
Dose	Up to 600 mcg (6 puffs)	2.5 mg	Up to 800 mcg (8 puffs)	5 mg
Max. single dose	600 mcg	2.5 mg	800 mcg	5 mg
Dosing interval	5-15 min PRN	5-15 min PRN	5-15 min PRN	5-15 min PRN
Max. # of doses	3	3	3	3

*1 puff=100 mcg

Bronchoconstriction - Treatment– ALS PCS 5.0

Consider EPINEPHrine	
	Route
	IM
	Concentration
	1 mg/mL = 1:1,000
Dose	0.01 mg/kg*
Max. single dose	0.5 mg
Dosing interval	N/A
Max. # of doses	1

*The EPINEPHrine dose may be rounded to the nearest 0.05 mg

Consider dexamethasone	
	Route
	PO/IM/IV
Dose	0.5mg/kg
Max. single dose	8mg
Dosing interval	N/A
Max. # of doses	1

Bronchoconstriction - Treatment

CLINICAL CONSIDERATIONS

Epinephrine should be the first drug administered if the patient is apneic. Salbutamol MDI may be administered subsequently using a BVM MDI adapter (if available).

Nebulization is contraindicated in patients with a known or suspected fever or in the setting of a declared febrile respiratory illness outbreak by the local medical officer of health.

When administering salbutamol MDI, the rate of administration should be 100 mcg approximately every 4 breaths.

A spacer should be used when administering salbutamol MDI (if available).

Crackles

- What is the underlying cause in children?
  - Pneumonia? Aspiration (especially in infants)? Bronchiolitis (RSV)?
- Oxygenation
- PPV prn

Respiratory Failure

e.g. severe asthma,

- PPV with O_2 **
- SpO2, $ETCO_2$
- allow for longer expiratory time (I:E ratio) if air trapping suspected
- IM epinephrine for asthma requiring PPV

Febrile Respiratory (enteric) Illness

- Decision to protect self based on suspected FREI
- PPE
- MDI with aerochamber (no nebulizer therapy)
  - ALS PCS contraindicates nebulized routes when fever present

Assessment Summary

- Assess ABCs and neuro
- Auscultate the chest for adventitious sounds
  - Lung sounds before and after Tx
- LOA before and after Tx
- Monitor colour before and after Tx
- SpO_2 on room air – then on O_2
- HR and ECG
  - HR high from hypoxia (slow is an ominous sign)
- ECG for irregularities
  - does ECG normalize with O_2 or other Tx

Treatment Summary

- Ensure proper sized equipment for each patient
- Least invasive → most invasive (i.e. supplemental O2, BVM, airways adjuncts….)
- Prepare and anticipate for declining status
- Know the differences (anatomy and physiology)
  - Proper technique and equipment will help prevent complications (i.e. anatomical obstructions)
- NIPPV – know correct Tv – overventilation can cause a decrease in CO

QUESTIONS

References

- Full article: Prehospital Pediatric Respiratory Distress and Airway Management Interventions: An NAEMSP Position Statement and Resource Document (tandfonline.com)
- ALS PCS 5.3